Quest 5: The Cartograffity

Lord of the Wings · growing land, sand and trees on a grid

> This guide uses the official puzzle text. The story is summarised and the rules are restated in plain words. It explains what the quest asks and the logic behind my solutions, not the code itself. Every number in the examples and traces was checked by running my solutions on the sample data (plus small instrumented copies to print intermediate values).

THE QUESTION
Start from the centre cell of a 0/1 grid, grow land outwards with a neighbour-count rule, and count cells of one kind.

INPUT
A grid of `0` (light) and `1` (dark) with an odd number of rows and columns.

OUTPUT
A count of cells: land (Part I), sand (Part II) or trees (Part III).

WHAT CHANGES BETWEEN PARTS
Each part keeps everything from the previous one and adds a layer: land → a sandy coastline on the land → trees on the land that is left.

The story

Legolink has joined the expedition, and the Grand Council hands over maps to Nulldor. The maps look like static: just light and dark patches. Legolink teaches everyone to read them: from the centre, the land "grows" over the dark patches, then a beach forms around its edge, and finally trees get stamped onto the solid ground with a potato stamp.

Understanding the question

Every rule looks at a cell's **eight neighbours** (including diagonals; fewer on the map's edge, only 3 in a corner). The sample is 11 rows × 21 columns, so its centre is row 5, column 10 (counting from 0).

Rule 1: growing land (all parts)

1. The centre cell is land `#`.
2. A cell that **touches the land** becomes land if at least **5** of its 8 neighbours are "good": land `#` or dark `1`.
3. Repeat until nothing changes.

A blob of `1`s that never touches the land stays water, however dark it is.

example

```
10#
1?0
110
```

touches # and has 5 good neighbours → becomes land

Rule 2: sand (Parts II and III)

1. Every land cell with a **water** neighbour (a `0`/`1` that didn't become land, or the map's edge) becomes sand `=`.
2. Then, repeatedly, a land cell with at least **5** sand neighbours also becomes sand.

Rule 3: trees (Part III)

Try land cells in reading order: a cell becomes a tree `@` if all 20 cells of the stamp around it are land or tree (a new tree counts straight away). Then the forest grows **twice**, each step **simultaneous**: a land cell with at least **4** tree neighbours and **no** sand neighbour becomes a tree.

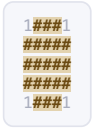
the 20-cell stamp

```
.###.
#####
##X##
#####
.###.
```

5×5 with corners cut off

Extra examples of my own (all run through my solution)

Map	Land	Sand	Trees	Why
5×5 all `0`	1	1	0	nothing grows; the centre touches water
5×5 all `1`	21	21	0	corners (3 neighbours) can't reach 5
9×9 all `1`	77	32	9	a 3×3 block of trees
11×11 all `1`	117	40	25	a 5×5 block of trees



5×5: land



9×9: final

Shared approach

All three parts grow the land exactly as in Part I; Parts II and III add their layer on top of that same land.

> **Fragile, not wrong:** the solution assumes plain newline line endings. A file saved with Windows line endings would make the grid look one column wider and move the "centre". Your input files are fine.

PART I How much land?

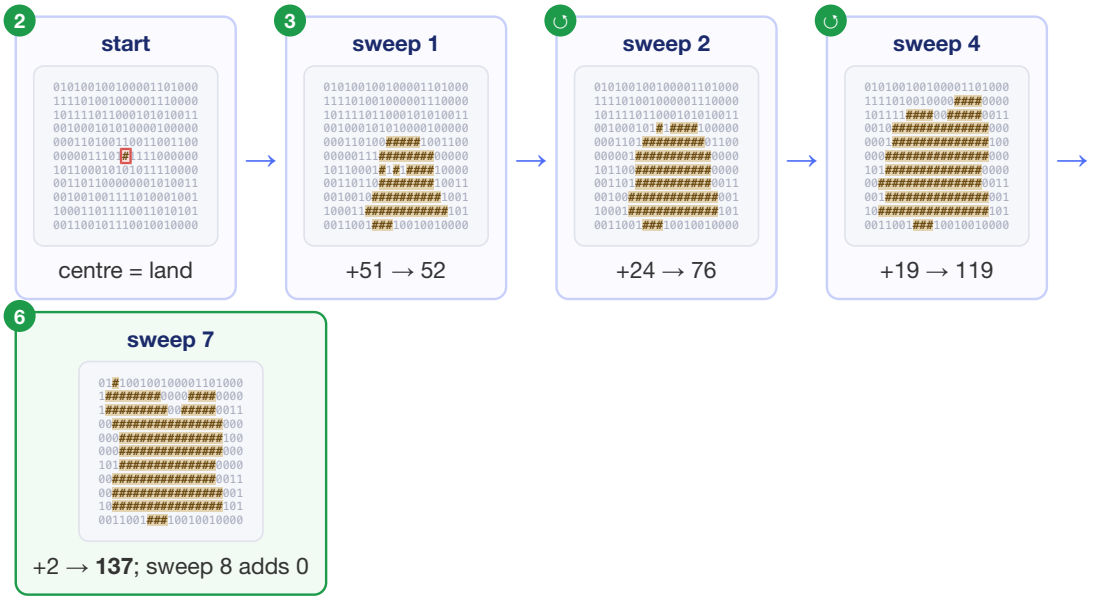
Grow the land from the centre (Rule 1) until it stops, then count the `#` cells.

Step-by-step solution

1. Read the grid of cells.
2. Mark the centre cell (row $\lfloor \text{rows}/2 \rfloor$, column $\lfloor \text{cols}/2 \rfloor$) as `#`.
3. Sweep over every cell that isn't land yet. Count its land neighbours and its good neighbours (land or `1`).
4. If it has at least 1 land neighbour and at least 5 good ones, make it `#`.
5. ↻ *Repeat the sweep until one sweep changes nothing.*
6. Count the `#` cells.

How my solution works

Sweep the whole grid in reading order, turning qualifying cells into land **straight away**, and repeat until a sweep changes nothing. The sample:



Dots match the step numbers above. The puzzle grows in 10 "all at once" rounds (4, 8, 17, … cells); my sweeps reach the same 137 cells in 7.

> **Why in-place still gives the puzzle's answer:** a new land cell only ever makes neighbours *more* likely to qualify, and land is never removed. So the order of additions can't change the final set: both reach the same **fixed point** of a **monotone** rule.

> **The map's edge:** the solution treats cells outside the map as not good. Because of this, a corner (3 neighbours) can never become land, and an edge cell needs all 5 of its neighbours to be good. That's how the 5×5 all-`1` example ends up with 21 land cells, not 25.

PART II How much sand?

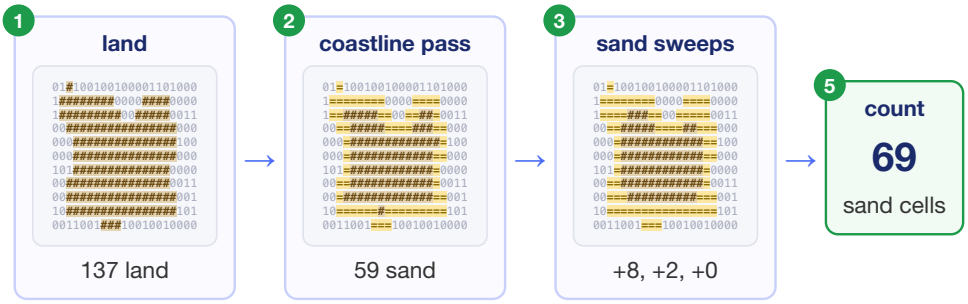
Grow the land as in Part I. Then turn the coastline into sand: first every land cell that touches water (or the map's edge), then repeatedly every land cell with at least 5 sand neighbours. Count the `=` cells.

Step-by-step solution

1. Do Part I to get the land.
2. Go over every land cell once. If any cell in its 3×3 block is off the map, or is `0` / `1`, make it `=`.
3. Sweep over the remaining land cells. If one has at least 5 `=` neighbours, make it `=`.
4. ↻ *Repeat until one sweep changes nothing.*
5. Count the `=` cells.

How my solution works

Grow the land as in Part I. Then one pass turns the coastline into sand (only `0` / `1` or the map edge count as water, so new sand doesn't spread inward in that same pass), and sweeps grow the sand until nothing changes.



Dots match the step numbers above. The puzzle's rounds add 6 + 3 + 1; my sweeps add 8 + 2. Both give 69.

PART III How many trees?

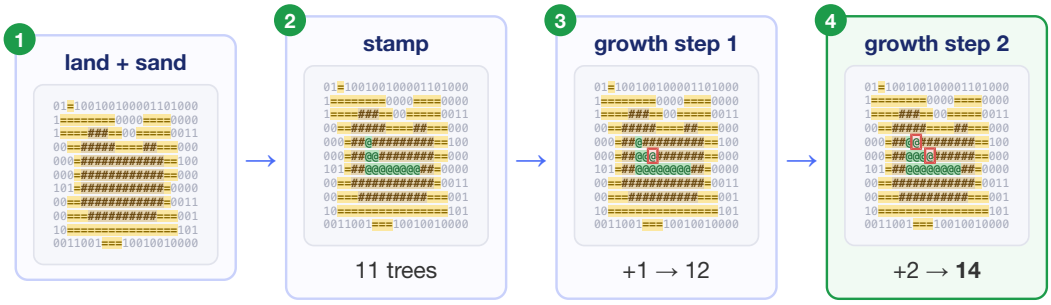
Build land and sand as before. Then stamp trees onto land cells in reading order, using the 20-cell stamp, and let the forest grow for exactly two simultaneous steps (at least 4 tree neighbours, no sand neighbour). Count the `@` cells.

Step-by-step solution

1. Do Part II to get the land and sand.
2. Go over land cells in reading order. If all 20 stamp cells around it are on the map and are `#` or `@`, make it `@` right away.
3. Growth step: collect every land cell that has no `=` neighbour and at least 4 `@` neighbours. Then turn all of them into `@` together.
4. Do the growth step a second time.
5. Count the `@` cells.

How my solution works

Stamp trees in reading order wherever all 20 stamp cells are land or tree (cells too near the edge can never pass). Then grow twice: each step first **collects** every land cell with no sand neighbour and ≥ 4 tree neighbours, and only then turns them all into trees.



Dots match the step numbers above; outlined cells are the new trees. Stamping in any order gives the same 11 trees, because a stamp only turns `#` into `@`, which still passes every other check.

Why growth must be simultaneous (my solution gets it right): in a copy where growth changes cells immediately, step 1 adds (5,8) through (5,12), each new tree giving the next cell its 4th tree neighbour, and the answer becomes **17** instead of 14. With a fixed number of steps, update order matters.

The map's edge again: the growth check skips off-map neighbours, so it doesn't see the "water around the map" there. That's safe: any land cell on the edge already became sand in Part II, so the only `#` cells left are inside the map.

Summary

Part	Counts	New rule	How cells are updated	Sample
I	land `#`	touches land and ≥ 5 neighbours are `#` / `1`	in place, repeat until no change	137
II	sand `=`	touches water/edge; then ≥ 5 `=` neighbours	coastline in one pass; then in place until no change	69
III	trees `@`	20-cell stamp; then ≥ 4 `@` and no `=` neighbours	stamp in place; growth simultaneous, exactly 2 steps	14

Cost: each sweep looks at every cell and its 8 neighbours (24 for the stamp), so it costs O(rows × cols). The land and sand stages repeat that until nothing changes (8 and 3 sweeps on the sample), which is fast enough for these maps.